
AFDBench: A Benchmark for Evaluating AI-Generated National Weather Service Forecast Discussions

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Abstract

Area Forecast Discussions (AFDs) are the highest-stakes scientific text produced by the U.S. National Weather Service—errors carry life-safety consequences. Despite rapid progress in AI weather prediction, no benchmark exists for evaluating whether language models can generate professional meteorological text. We introduce **AFD-Bench**, a benchmark comprising 7,732 expert-written AFDs from 13 NWS offices spanning 4 months and 7 U.S. climate regions, each paired with structured AI weather forecast data from Google’s WeatherNext 2. AFDBench defines three complementary evaluation metrics: *Met-Align* (numerical accuracy against human reference), *Style-Align* (adherence to NWS professional vocabulary), and *Input-Grounding* (fidelity to source weather data). We establish baselines with three open-source LLMs (7–8B parameters), finding that all achieve ~13% Met-Align, ~0.33 Style-Align, and ~0.88 Input-Grounding in zero-shot evaluation—confirming a significant gap between LLM capabilities and professional meteorological writing. We release AFDBench as a public benchmark to accelerate research on scientific text generation where numerical precision and domain expertise are essential.

1. Introduction

The National Weather Service (NWS) issues Area Forecast Discussions (AFDs) from 122 Weather Forecast Offices (WFOs) across the United States, providing the scientific reasoning behind weather forecasts. Unlike automated forecast products that convey *what* will happen, AFDs explain

why—detailing synoptic-scale patterns, mesoscale interactions, model disagreements, and forecaster confidence. AFDs are read by emergency managers, pilots, media, agriculture, and the general public; numerical errors or reasoning failures can have life-safety consequences.

Recent breakthroughs in AI weather prediction—GraphCast (Lam et al., 2023), Pangu-Weather (Bi et al., 2023), GenCast (Price et al., 2025)—have achieved human-competitive skill at predicting gridded atmospheric fields. However, these systems produce numerical arrays, not text. The critical “last mile” of weather communication—translating model output into expert-level natural language—has received no systematic evaluation.

Large language models (LLMs) are increasingly capable text generators, but their application to meteorological writing faces unique challenges: (1) AFDs require **numerical precision**—temperature values, wind speeds, and pressure readings must be grounded in data, not hallucinated; (2) AFDs demand **domain-specific vocabulary**—terms like “vorticity advection,” “lee cyclogenesis,” and “moisture-starved boundary” have precise meteorological meanings; (3) AFDs follow **structural conventions**—section headers (.SHORT TERM, .AVIATION), delimiters (&&), and temporal framing that are unique to NWS products.

No existing benchmark tests these capabilities. General-purpose text generation benchmarks (e.g., summarization, translation) do not evaluate numerical grounding or domain vocabulary. Scientific QA benchmarks test factual recall, not text generation. Weather-specific AI benchmarks (WeatherBench 2) evaluate gridded predictions, not text.

We introduce **AFDBench** to fill this gap. Our contributions are:

1. **A curated dataset** of 7,732 professional AFDs from 13 NWS offices, each paired with structured WeatherNext 2 forecast data, in a reasoning-first 4-field format.
2. **Three evaluation metrics** that capture distinct dimensions of meteorological text quality: numerical accuracy (Met-Align), professional style (Style-Align), and data fidelity (Input-Grounding).

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3. **Comprehensive baselines** establishing the current capability gap between open-source LLMs and professional meteorologists.
4. A **geographic hold-out protocol** for evaluating whether models learn generalizable meteorological reasoning or station-specific patterns.

2. Related Work

AI Weather Prediction. GraphCast (Lam et al., 2023), Pangu-Weather (Bi et al., 2023), and GenCast (Price et al., 2025) predict gridded atmospheric state variables. WeatherBench 2 provides standardized evaluation for these numerical predictions. AFDBench addresses the complementary challenge of evaluating AI-generated meteorological *text*.

Scientific Text Benchmarks. SciQ, SciFact, and related benchmarks evaluate scientific question answering and claim verification. Clinical report generation benchmarks (Liu et al., 2019) evaluate medical text, the closest domain analog. No prior benchmark targets meteorological text generation, which uniquely requires joint numerical precision, domain vocabulary, and structured formatting.

Domain-Specific NLG Evaluation. Metrics for domain-specific text generation include BLEU, ROUGE, and BERTScore for general quality, plus domain-specific metrics such as clinical accuracy scores for medical reports. AFDBench introduces metrics tailored to meteorological requirements: numerical set overlap (Met-Align), domain vocabulary coverage (Style-Align), and input data fidelity (Input-Grounding).

3. Dataset Construction

3.1. Source Data: NWS Area Forecast Discussions

AFDs are archived by the Iowa Environmental Mesonet (IEM) and publicly available. Each AFD is a free-text document typically 500–4,000 words long, organized into sections (`.SHORT TERM`, `.LONG TERM`, `.AVIATION`, etc.) separated by `&&` delimiters. AFDs are issued 2–4 times daily by each WFO.

We collected AFDs from **13 NWS offices** (Table 1) spanning 7 U.S. climate regions, covering **December 31, 2025 through April 6, 2026** (97 unique dates). This winter-to-spring window captures diverse synoptic regimes: winter storms, cold air outbreaks, lee cyclogenesis, early-spring severe weather, and marine-influence patterns.

3.2. Weather Data: WeatherNext 2 Integration

Each AFD is paired with a structured JSON object containing a single-timestep forecast from Google’s WeatherNext 2

Table 1. AFDBench station network. 11 offices for training, 2 held out for geographic evaluation. Sample counts reflect the number of successfully paired AFD + WeatherNext 2 records.

WFO	City	<i>n</i>	Region	Split
OKX	New York, NY	888	Northeast	Train
LOX	Los Angeles, CA	760	West Coast	Train
MTR	San Francisco, CA	740	West Coast	Train
LMK	Louisville, KY	690	Ohio Valley	Train
PHI	Philadelphia, PA	652	Mid-Atlantic	Train
BOU	Denver, CO	581	Mountain	Train
TOP	Topeka, KS	589	Central Plains	Train
DMX	Des Moines, IA	579	Upper Midwest	Train
EWX	Austin/San Antonio	428	South Central	Train
FWD	Dallas/Fort Worth	399	South Central	Train
SEW	Seattle, WA	393	Pacific NW	Train
BOX	Boston, MA	557	New England	Test
MRX	Morristown, TN	476	SE Appalachian	Test
Total		7,732		

AI weather prediction system, matched by station and valid time. The input includes:

- **Surface conditions:** temperature (F/K), wind speed (mph/kt) and direction, relative humidity, mean sea-level pressure (hPa/inHg), 6-hour precipitation.
- **Comfort indices:** heat index, wind chill, apparent temperature.
- **Upper-air fields:** 850 mb and 500 mb temperature and wind, 1000–500 mb thickness.
- **Ensemble spread:** temperature spread (K) indicating forecast uncertainty.

This pairing creates a realistic evaluation setting: the model must interpret AI weather data and compose expert text, rather than regurgitating values from the target. WeatherNext 2 forecasts were retrieved from Google Cloud Storage (Zarr format) at the nearest grid point to each WFO, using the forecast cycle closest to the AFD issuance time with a 6-hour lead time.

3.3. Reasoning-First Data Format

Each sample is stored in a 4-field JSONL format designed to elicit structured meteorological reasoning:

1. **Instruction:** Task prompt specifying the NWS office (e.g., “Act as an NWS Meteorologist for DMX. Generate a Forecast Discussion.”)
2. **Input:** Structured WeatherNext 2 JSON (mean: 56 words per sample).

3. **Thinking:** Synoptic reasoning extracted from the AFD’s synopsis section—the meteorological “chain of thought” (mean: 57 words, range: 8–180).
4. **Output:** The complete professional AFD text (mean: 951 words, range: 151–3,952).

The Thinking field captures the *why* behind the forecast—the synoptic analysis that professional meteorologists perform before writing the discussion. This structure enables training and evaluating models on both the reasoning process and the final output.

3.4. Data Split

We adopt a **geographic hold-out** protocol: two offices—BOX (Boston, MA) and MRX (Morristown, TN)—are completely excluded from training and reserved for evaluation. These offices represent climatologically distinct regions (maritime Northeast vs. Appalachian Southeast) not represented in the training distribution, testing whether models learn generalizable meteorological reasoning or station-specific patterns.

The split yields **6,699 training samples** (11 offices) and **1,033 test samples** (2 offices). We chose geographic hold-out over random splitting because random splits would leak station-specific patterns (seasonal norms, terrain effects, local terminology) from train to test.

4. Evaluation Metrics

AFDBench defines three metrics that capture distinct dimensions of meteorological text quality. All metrics are fully automated, requiring no human annotation.

4.1. Met-Align (%): Numerical Accuracy

Met-Align measures the overlap of multi-digit numbers between the generated and human reference texts:

$$\text{Met-Align} = \frac{|\mathcal{N}_{\text{ai}} \cap \mathcal{N}_{\text{ref}}|}{|\mathcal{N}_{\text{ref}}|} \times 100 \quad (1)$$

where $\mathcal{N}$ denotes the set of 2–3 digit numbers extracted via regex `(\b\d{2,3}\b)`. Human AFDs contain a mean of 33.4 such numbers (median: 25), covering temperatures, wind speeds, precipitation amounts, pressure values, and time references. Met-Align of 100% indicates perfect numerical agreement with the human expert; 0% indicates complete hallucination.

Limitations. Met-Align uses set intersection, ignoring semantic context: a wind speed “25” matching a temperature “25” counts as a hit. Additionally, numbers like “12” (an hour) and “12” (a temperature) are conflated. Despite

these limitations, Met-Align provides a tractable proxy for numerical faithfulness that correlates with meteorological accuracy.

4.2. Style-Align (0–1): Professional Vocabulary

Style-Align measures adherence to the NWS professional register via overlap of domain vocabulary tokens:

$$\text{Style-Align} = \frac{|\mathcal{V}_{\text{ai}} \cap \mathcal{V}_{\text{ref}}|}{|\mathcal{V}_{\text{ref}}|} \quad (2)$$

where $\mathcal{V}$ is the set of NWS domain terms present in the text. The vocabulary set comprises 12 canonical meteorological terms: SYNOPSIS, DISCUSSION, AVIATION, MARINE, CONVECTION, TROUGH, RIDGE, FRONTAL, PRECIPITATION, MESOSCALE, VORTICITY, ADVECTION. Human AFDs use a mean of 4.1 of these terms per discussion (median: 4).

Style-Align captures whether the model has learned the professional meteorological register. A model generating fluent but generic weather text (“it will be sunny and warm”) scores low; one using NWS-standard terminology (“subsidence under upper ridge will limit convective development”) scores high.

4.3. Input-Grounding (0–1): Data Fidelity

Input-Grounding measures whether the model faithfully uses the WeatherNext 2 input data rather than hallucinating plausible-sounding values. It evaluates three independently scored conditions:

1. **Temperature grounding:** Does any number in the output fall within 3°F of the input surface temperature? (1.0 if $\leq 3^\circ\text{F}$, 0.5 if $\leq 10^\circ\text{F}$, 0.0 otherwise.)
2. **Wind direction:** Does the correct cardinal wind direction (e.g., “NW”) from the input appear in the output? (1.0 if present, 0.0 otherwise.)
3. **Pressure regime:** If MSLP > 1020 hPa, does the text mention “high” or “ridge”? If MSLP < 1010 hPa, does it mention “low” or “trough”? (1.0 if correct, 0.5 for ambiguous range, 0.0 if wrong.)

Input-Grounding is the mean of these three checks. Unlike Met-Align (which compares to the human reference), Input-Grounding compares to the *input data*, measuring whether the model correctly interprets its weather inputs regardless of whether the human chose the same framing.

4.4. Metric Independence

The three metrics are designed to be complementary:

- A model that copies all numbers from a database scores high on Met-Align but low on Style-Align (no professional vocabulary) and possibly low on Input-Grounding (numbers may not match the specific input).
- A model that writes beautiful NWS prose with hallucinated numbers scores high on Style-Align but low on Met-Align and Input-Grounding.
- A model that correctly interprets its input but uses informal language scores high on Input-Grounding but low on Style-Align.

Human NWS experts achieve approximately 1.0 on all three metrics by construction.

5. Baseline Evaluation

5.1. Experimental Setup

We evaluate three open-source instruction-tuned LLMs in zero-shot mode:

- **Qwen2.5-7B-Instruct** (Hui et al., 2024): 7B parameters.
- **Hermes-3-Llama-3.1-8B**: 8B parameters, instruction-tuned Llama 3.1.
- **Mistral-7B-Instruct-v0.3**: 7B parameters.

Each model receives the instruction, input (WeatherNext 2 JSON), and a generation prompt. We use greedy decoding (temperature 0, no sampling) with a maximum of 512 new tokens. All models are evaluated in fp16 precision on a single GPU.

Zero-shot baselines are evaluated on a 200-sample subset of the test set. We additionally evaluate a GRPO-trained model (AI-Met-v1) on the full 1,033-sample test set to demonstrate the benchmark’s sensitivity to training interventions.

5.2. Results

Table 2 presents the baseline results on the multi-timestep v3 version of AFDBench.

Multi-Timestep Inputs Break the Met-Align Ceiling. Under the original single-timestep constraint (v1), models were strictly capped at a Met-Align ceiling of $\sim 14\%$ because human AFDs naturally synthesize multiple future periods (e.g., today’s high, tonight’s low, and days 3–7). Transitioning to multi-timestep forecast inputs (v3) solves this limitation. The multi-timestep SFT and GRPO models achieve Met-Align scores of 19.91% and 18.68% respectively, showing a massive relative improvement over the zero-shot baseline (11.32%) and the single-timestep floor.

Table 2. AFDBench baseline results on multi-timestep inputs (v3). All zero-shot models show a significant gap to human performance. Multi-timestep SFT and GRPO models successfully break the $\sim 14\%$ Met-Align floor of single-timestep models. *Evaluated on 50-sample subset.

Model	Params	Met	Style	Ground
Human NWS Expert	—	100.0	1.000	1.000
AI-Met-v3 (GRPO)	7B	18.68*	0.463*	0.597*
AI-Met-v3 (SFT)	7B	19.91*	0.433*	0.587*
Qwen2.5-7B-Inst. (Zero-Shot)	7B	11.32*	0.329*	0.847*
Mistral-7B-v0.3 (Zero-Shot)	7B	12.95*	0.327*	—
Hermes-3-Llama (Zero-Shot)	8B	12.84*	0.338*	—

Style-Align Reveals the Professional Register. Zero-shot Style-Align scores of ~ 0.33 indicate that models reproduce only one-third of the NWS professional register. Fine-tuning models with SFT increases this register to 0.433. Applying GRPO reinforcement learning further pushes the Style-Align metric to 0.463, indicating that reinforcement learning effectively aligns models to use domain-specific meteorological vocabulary.

Input Grounding Performance. The zero-shot Qwen baseline achieves 0.847 on Input-Grounding, which drops to 0.587–0.597 for the SFT and GRPO models. This drop indicates that while fine-tuned models generate highly realistic, domain-specific discussions that are well-aligned with the reference discussions stylistically, they occasionally hallucinate values or construct narratives that deviate from the strict numerical constraints of the input database, showing that grounding remains a major open research problem.

6. Dataset Analysis

6.1. Output Length and Complexity

Human AFDs in AFDBench range from 151 to 3,952 words (mean: 951). Shorter AFDs typically correspond to quiescent weather periods (“no significant weather expected”), while longer AFDs arise during active weather patterns requiring detailed reasoning about multiple synoptic features, mesoscale interactions, and forecast uncertainty. The length distribution is right-skewed: most AFDs are 500–1,200 words, with a long tail extending beyond 3,000 words during complex events.

6.2. Numerical Density

Human AFDs contain a mean of 33.4 multi-digit numbers per discussion (median: 25), covering temperature forecasts, wind speeds, precipitation amounts, pressure values, geographic coordinates, and time references. This high numerical density makes AFD generation a stringent test of numerical grounding—a model must produce dozens of

contextually appropriate numbers per sample.

6.3. Station Variability

Sample counts range from 393 (SEW/Seattle) to 888 (OKX/New York), reflecting differences in AFD issuance frequency and the collection window. Coastal offices (OKX, LOX, MTR) tend to produce more AFDs due to marine-influenced weather requiring frequent updates. Mountain offices (BOU) produce detailed AFDs with complex terrain-driven discussions.

6.4. Temporal Coverage

The 97-day collection window (Dec 31, 2025–Apr 6, 2026) captures the winter-to-spring transition, including: (1) deep winter cold air outbreaks and nor’easters (January–February), (2) spring transition with temperature volatility and convective initiation (March), (3) early spring severe weather setups (late March–April). This window excludes peak tornado season (May–June) and peak wind season (November); extending coverage is a priority for future work.

7. Limitations and Future Work

Multi-Timestep Inputs. While the initial release of AFD-Bench (v1) was constrained by single-timestep inputs, we have addressed this in the current version (v3) by providing multi-timestep forecast sequences covering the full AFD time horizon. However, future extensions will incorporate even more granular spatial grids and intermediate cycles to capture rapid weather evolutions.

U.S. Only. AFDBench covers only U.S. NWS offices. International weather services (Met Office, DWD, JMA) use different formats, terminology, and languages. Multilingual meteorological benchmarks would test cross-cultural transfer.

13 of 122 Offices. We cover 13 of 122 NWS WFOs. While selected for geographic and climatic diversity, this is a sample. Tropical offices (e.g., GUM/Guam, HFO/Honolulu), Alaska offices, and fire-weather-focused offices present distinct challenges not represented here.

Metric Coarseness. Met-Align uses bag-of-numbers comparison without semantic context. Style-Align uses a fixed 12-term vocabulary. Input-Grounding checks only three conditions. All three metrics are necessary but not sufficient for assessing AFD quality; human expert evaluation remains the gold standard.

No Semantic Evaluation. Current metrics do not evaluate whether the model’s *reasoning* is meteorologically sound—

whether it correctly identifies synoptic features, applies the right physical mechanisms, or appropriately conveys forecast uncertainty. LLM-as-judge approaches or expert annotation could address this gap.

Planned Extensions.

1. Multi-timestep WeatherNext 2 input (forecast hours 6, 12, 24, 48, 72, 120, 168).
2. Expand to 30+ offices covering all U.S. climate zones.
3. Year-round coverage including summer convective season and autumn wind season.
4. Human expert evaluation protocol with operational NWS forecasters.
5. Retrieval-augmented baselines using historical AFDs.
6. Larger proprietary model baselines (GPT-4o, Claude, Gemini).

8. Broader Impact

AFDBench is built from publicly available NWS data and AI weather forecasts. We note several considerations:

Not for Operational Use. AFDBench is a research benchmark. AI-generated AFDs should not be used for operational weather communication without expert review. NWS forecasters apply judgment, local knowledge, and situational awareness that cannot be captured by automated metrics.

Potential Benefits. A capable AI meteorologist could assist human forecasters by generating first drafts, flagging unusual patterns, or maintaining consistency across shift changes. This could improve forecast communication in resource-constrained offices.

Risks. Overconfidence in AI-generated meteorological text could lead to missed hazards or delayed warnings. The Met-Align metric’s coarseness means a high-scoring model may still contain critical errors. Any deployment must include human oversight.

9. Conclusion

We introduced AFDBench, the first benchmark for evaluating AI-generated meteorological text. By pairing 7,732 expert-written NWS Area Forecast Discussions with real AI weather forecast data and defining three complementary metrics, AFDBench enables systematic evaluation of whether language models can bridge the gap between numerical weather prediction and expert weather communication.

Our baselines reveal that current open-source LLMs achieve only $\sim 33\%$ of human professional vocabulary usage and $\sim 88\%$ input data fidelity, establishing clear targets for future work. We release AFDBench to the research community to accelerate progress on scientific text generation where numerical precision and domain expertise are essential.

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A. Sample Data Record

The following figure shows a complete AFDBench data record, illustrating the 4-field reasoning-first format with WeatherNext 2 input.

B. Metric Implementation

All three metrics are implemented in Python using only the `re` and `json` standard library modules, with no external dependencies. The evaluation script processes $\sim 1,000$ samples in under 1 second on a standard CPU (metric computation only, excluding model inference). Source code is included with the dataset release.

Instruction: Act as an NWS Meteorologist for MRX. Generate a Forecast Discussion.

Input (WeatherNext 2 JSON):

```
{
  "station": "MRX",
  "valid_time": "2025-12-31T00:15Z",
  "surface": {
    "temperature_F": 26.4,
    "wind_speed_mph": 5.8,
    "wind_direction": "W",
    "relative_humidity_pct": 67.7,
    "mslp_hPa": 1022.2,
    "precipitation_6hr_in": 0.0,
    "upper_air": {
      "850mb_temperature_F": 19.9,
      "500mb_wind_direction": "NW",
      "1000_500mb_thickness_m": 4073.5,
      "ensemble": {
        "temperature_spread_K": 0.59
      }
    }
  }
}
```

Thinking: High pressure centered over the region with clear skies and light winds. Cold air in place with thickness values well below 5400m. No precipitation expected through the period.

Output: .SHORT TERM /THROUGH TONIGHT/. . . [Full professional AFD text, typically 500–2000 words]

Figure 1. Complete AFDBench data record from MRX (Morristown, TN). The WeatherNext 2 input provides surface, upper-air, and ensemble data; the Thinking field contains extracted synoptic reasoning; the Output is the full human-written AFD.